A BOUNDED-ABOVE COMPLEX OF FINITE STABLY FREE MODULES LIFTS ALONG AN IDEAL OF INVERTIBLE ELEMENTS

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Editorial note. The counted statement span contains a four-item enumerated hypothesis list; it is delivered verbatim, including the author's penultimate item about every element of 1 + I being invertible and the closing freeness sentence. The counted proof span refers to the hypotheses as properties (1) through (5) while the statement lists only four items; that numbering is delivered verbatim with no attempt to reconcile it. The five references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The abstract lifting lemma for complexes with respect to a class of modules closed under the five listed properties. invoked at proof lines 19532-19534. Produces the complex P^bullet of finite stably free modules.
- The class of finite stably free modules is an exact category, and finite stably free modules lift over an ideal consisting of elements of 1 + I. invoked at proof lines 19535 and 19538-19540. Supplies properties (2) and (4) of the lifting lemma.
- A pseudo-coherent complex is represented by a bounded above complex of finite free modules. invoked at proof lines 19541-19542. Supplies property (5) of the lifting lemma.
- Nakayama's lemma and the classification of isomorphism classes of finite projective lifts. invoked at proof lines 19536-19537 and 19543-19544. Give property (3) and the final freeness assertion.

Source reference. Original lemma 00DV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 512–512. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 70: sources/stacks-more-algebra/pinned-original-source.tex, source lines 70–70. Same-chapter navigation pointer; the statement it

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names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 145: sources/stacks-more-algebra/pinned-original-source.tex, source lines 145–145. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 19390: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19390–19390. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 93: sources/stacks-more-algebra/pinned-original-source.tex, source lines 93–93. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let $I \subset R$ be an ideal. Let $E^\bullet$ be a complex of R/I-modules. Let K be an object of $D(R)$. Assume that*

1. *$E^\bullet$ is a bounded above complex of finite stably free R/I-modules,*
2. *$K \otimes^{\mathbf{L}}_R R/I$ is represented by $E^\bullet$ in $D(R/I)$,*
3. *$K^\bullet$ is pseudo-coherent, and*
4. *every element of $1 + I$ is invertible.*

Then there exists a bounded above complex $P^\bullet$ of finite stably free R-modules representing K in $D(R)$ such that $P^\bullet \otimes_R R/I$ is isomorphic to $E^\bullet$. Moreover, if E^i is free, then P^i is free.

Proof. We apply Lemma 19390 using the class $\mathcal{P}$ of all finite stably free R-modules. Property (1) of the lemma is immediate. Property (2) follows from Lemma 70. Property (3) follows from Nakayama's lemma (Algebra, Lemma 00DV). Property (4) follows from the fact that we can lift finite stably free R/I-modules to finite stably free R-modules, see Lemma 93. Part (5) holds because a pseudo-coherent complex can be represented by a bounded above complex of finite free R-modules. The final assertion of the lemma follows from Lemma 145. □